Water Potability Predictor: Safeguarding Public Health with AI

AI & Society Case Study Project

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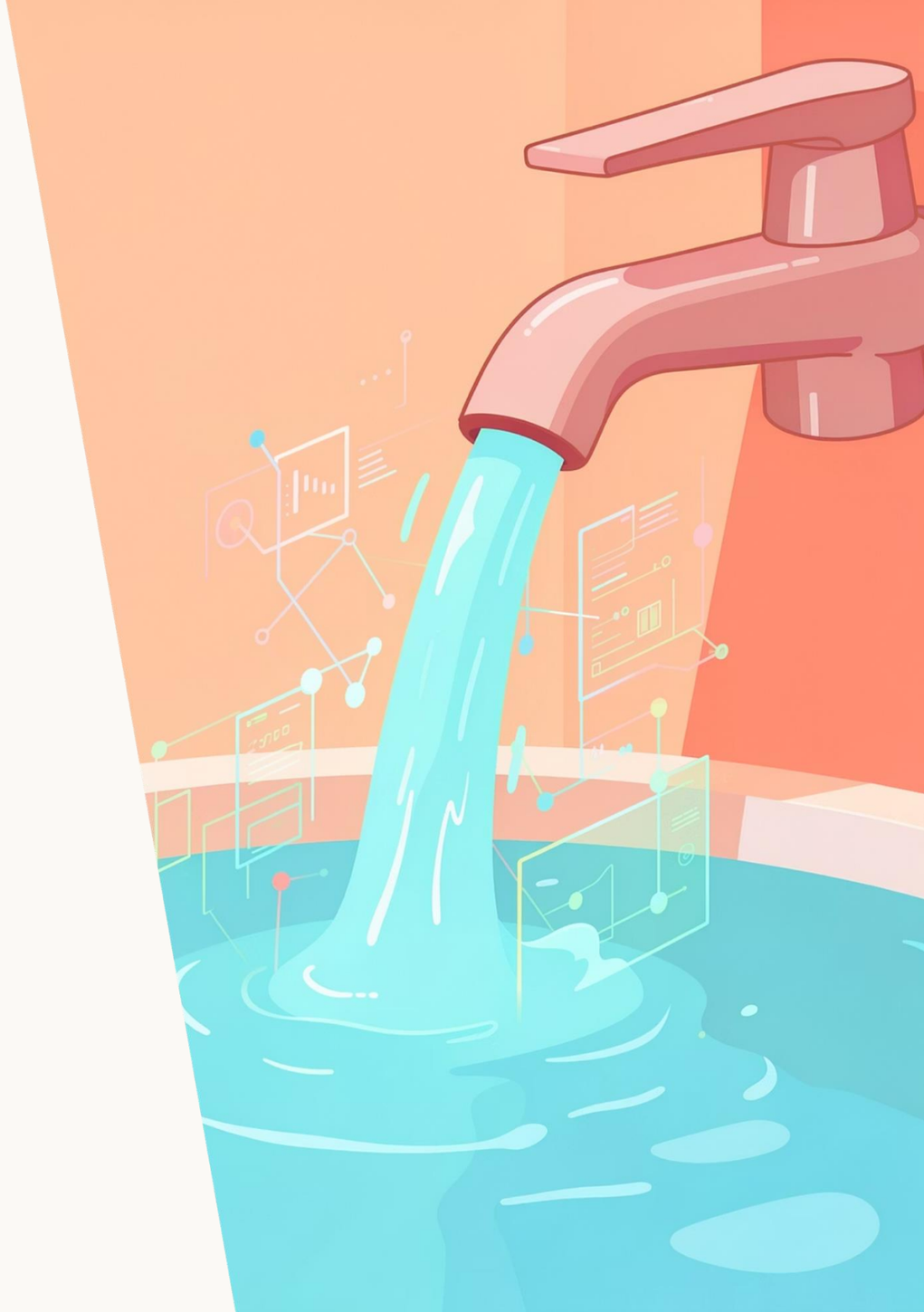

Project Division of Labor

Dataset Sourcing

Exploratory data analysis and data preprocessing

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Model Development

Machine learning model selection, training, and evaluation

[e23cseu1863],[e23cseu1862]

Backend Integration

Exporting model assets using Joblib and Scalers

[e23cseu1894],[e23cseu1869]

UI Development

Streamlit dashboard creation

[e23cseu1862],[e23cseu1865]

Impact Analysis

Societal impact analysis, ethical framework design, and final report compilation

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The Real-World Problem

Reactive vs Proactive

Traditional lab testing detects contamination only after it happens

Time-Consuming

Standard water tests take 24 to 72 hours for results

High Costs

Continuous lab testing is expensive and impractical for developing regions

Dynamic Ecology

Water quality fluctuates rapidly due to industrial waste, weather, and drainage

Aligning AI with Global Goals

Core Objective

Instantly classify water as "Drinkable" or "Not Drinkable" using AI

UN SDGs

Directly aligns with Goal 6 (Clean Water & Sanitation) and Goal 3 (Good Health)

AI Concept

Supervised Machine Learning (Classification) to find non-linear relationships in water chemistry

Human Limitation

AI spots complex chemical correlations (e.g., pH drops + high conductivity) that humans miss

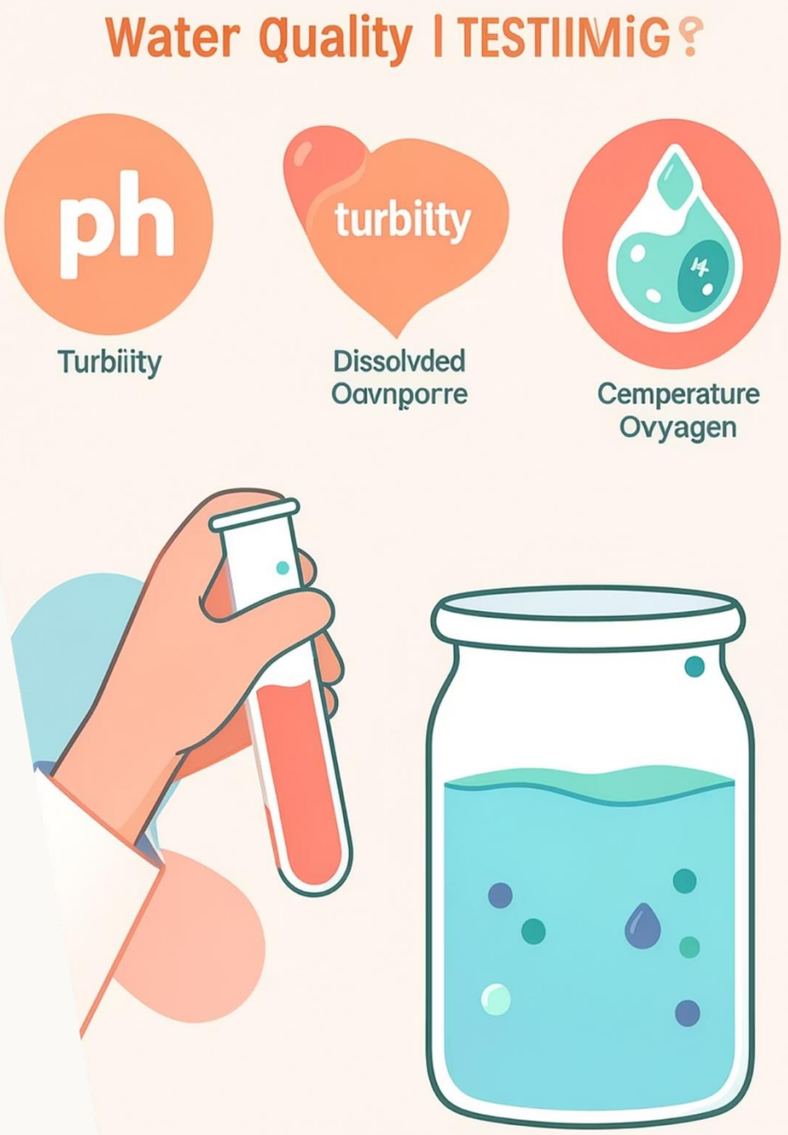
Kaggle Water Potability Dataset

Dataset Overview

- Source: Open-source Kaggle dataset
- Size: 3,276 water samples
- Features: 9 critical physical/chemical parameters
- Target Variable: Potability (1 = Safe, 0 = Unsafe)

Key Metrics

- pH, Hardness, Solids (TDS)
- Chloramines, Sulfates
- Conductivity, Organic Carbon
- Trihalomethanes, Turbidity



Step-by-Step Data Pipeline

Data Ingestion

Loading the 3,276 samples

Preprocessing

Handling missing values and scaling features using StandardScaler

Model Training

Training multiple algorithms (Logistic Regression, RandomForest, Support Vector Machine)

Exporting Assets

Saving the best model and scaler via Joblib

Deployment

Integrating the assets into a local Streamlit web application

Tools, Technologies & Algorithms

Algorithm & Backend

- **Support Vector Machine (SVC with RBF Kernel)**
- Excellent at handling complex, non-linear environmental data relationships
- Python, Scikit-Learn, Pandas, NumPy
- Joblib for .pkl file generation

Frontend & Deployment

- **Streamlit** for user interface
- Interactive web application
- Model persistence through file generation

Model Performance

Primary Metric Focus

Recall (Sensitivity) for the "Unsafe" class

Accuracy

Model achieved high accuracy in predicting decision boundaries

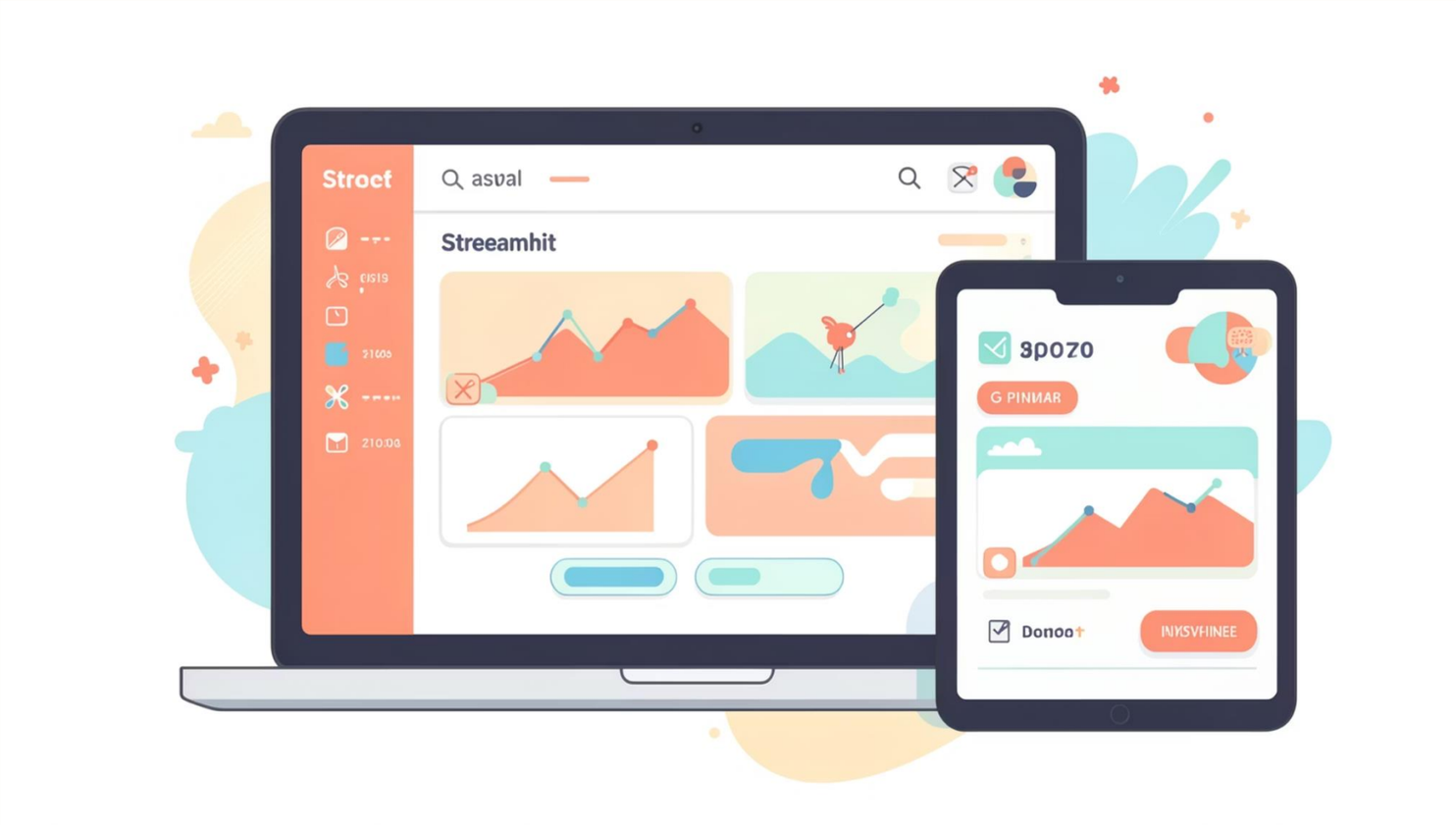
Analysis

The model successfully learned the threshold combinations of the 9 chemical metrics to determine potability

Preprocessing Impact

Feature scaling was crucial, as values ranged drastically (e.g., pH 0-14 vs. Solids up to 60,000)

How the System Works in Reality



1 User Interface

A simple, interactive Streamlit dashboard

2 Input Phase

User enters 9 water metrics obtained from basic testing strips or sensors

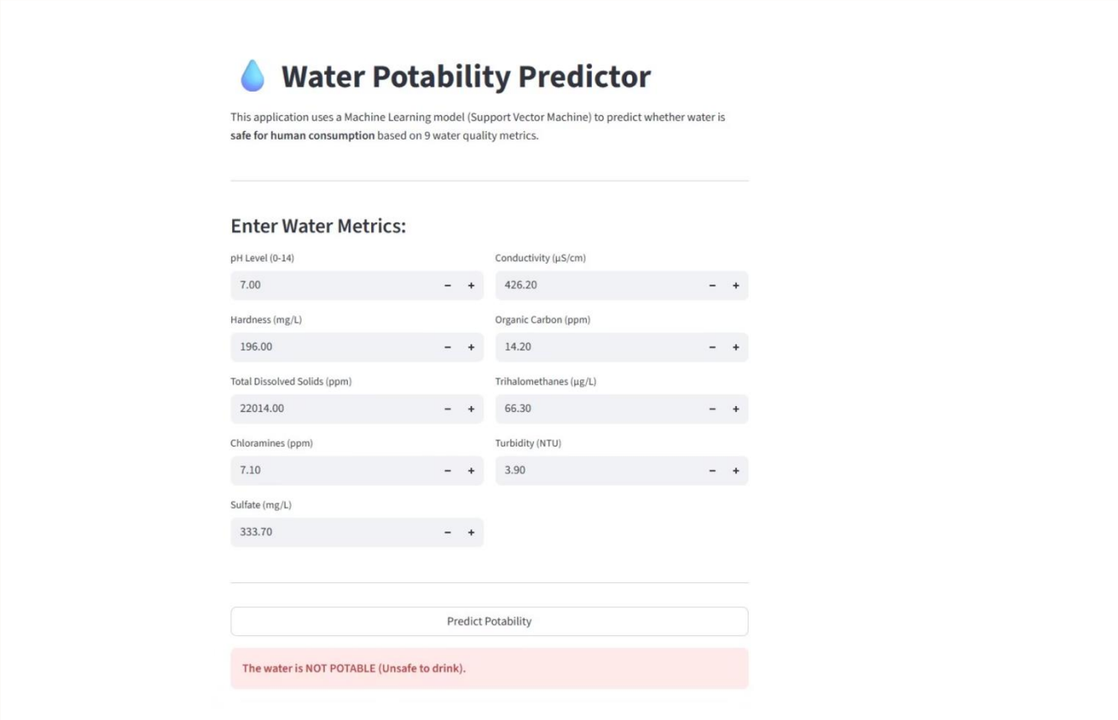
3 Processing

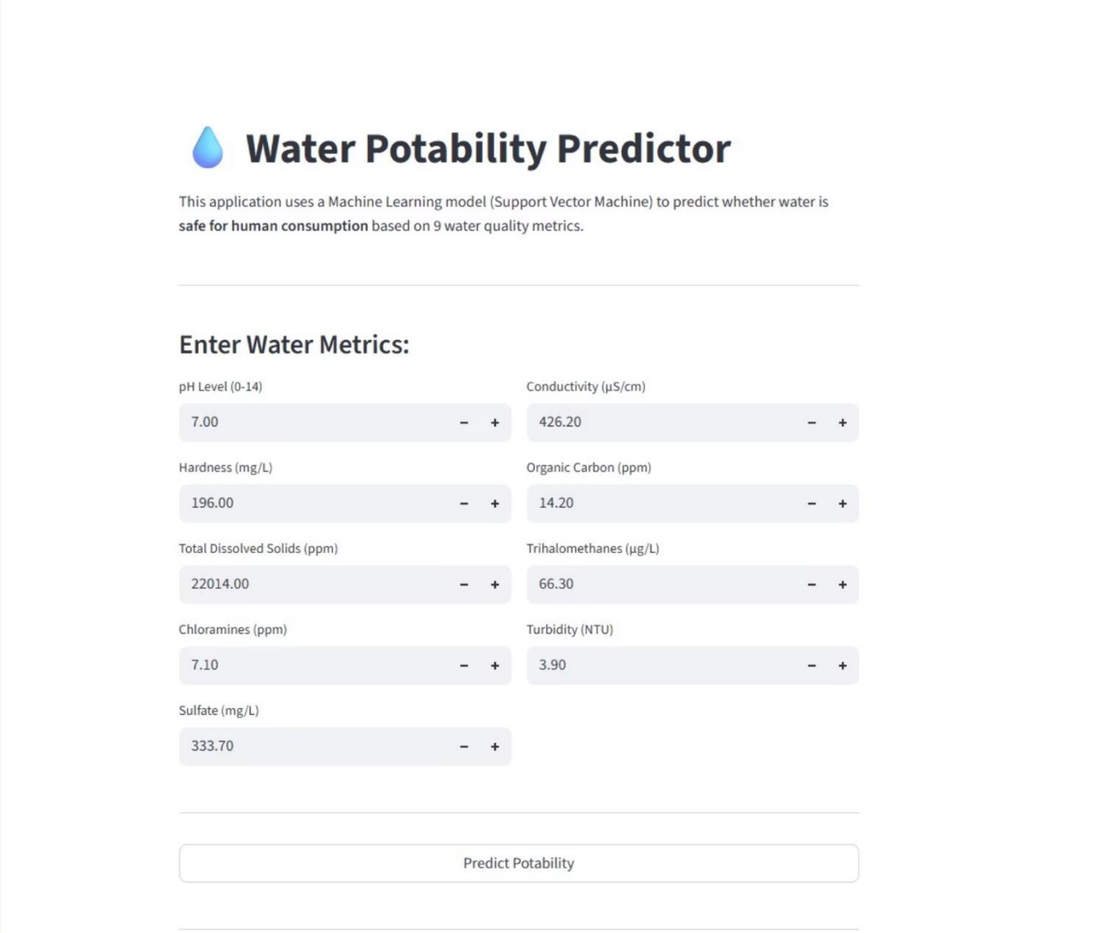
The app scales the user input behind the scenes using scaler.pkl

4 Output

Instant visual feedback predicting if the water is safe for human consumption

Application Interface in Action





System Evaluation

Advantages

- Delivers instant results compared to 72-hour lab waits
- Highly cost-effective, accessible, and acts as an early warning system

Limitations

- Cannot detect biological contaminants (e.g., bacteria, viruses)
- Relies on the user having access to basic chemical sensors to get the 9 input values
- Potential for "data drift" if local water chemistry changes over time

Transforming Public Health

Proactive Healthcare

Shifts society from treating waterborne diseases to preventing them. Early detection prevents outbreaks, reduces hospitalizations, and saves lives in vulnerable populations. By identifying risk before symptoms spread, communities can act faster, protect children and older adults, and stop small contamination events from becoming large-scale public health emergencies.

Democratizing Tech

Gives rural and underfunded communities access to top-level safety assessments. No longer dependent on expensive lab infrastructure or government resources that may not reach remote areas. This levels the playing field for communities that have historically been left behind, making high-quality water monitoring more equitable, scalable, and available where it is needed most.

Resource Optimization

Allows governments to deploy expensive lab resources only when the AI flags a high-risk area. Maximizes efficiency of public health budgets and enables data-driven decision-making at scale. Instead of testing everywhere all the time, officials can prioritize the highest-risk locations, respond more strategically, and stretch limited resources farther without sacrificing public safety.

The Ethics of AI in Healthcare

The "False Alarm" (False Positive)

Labeling safe water as unsafe can create inconvenience, confusion, and mistrust in the system. People may boil water unnecessarily, discard perfectly usable supplies, or turn to alternative sources that may be less convenient or more expensive. While this type of error does not directly harm health, it still carries ethical weight: repeated false alarms can erode confidence in the tool, burden families with avoidable effort, and make communities more likely to ignore future warnings—even when those warnings are truly urgent.

The "Missed Detection" (False Negative)

Labeling toxic water as safe is a life-threatening failure with potentially irreversible consequences. Families could consume contaminated water and be exposed to waterborne diseases, severe illness, long-term health complications, or death, especially among children, older adults, pregnant people, and immunocompromised individuals. Ethically, this is the most serious kind of error because it violates the core responsibility of healthcare systems: to protect people from preventable harm. A single missed detection can have consequences far beyond the individual household, spreading disease and undermining public trust in health interventions.

Our Ethical Choice

We deliberately prioritized Recall to minimize missed detections, because in healthcare and public safety, failing to warn people about danger is more serious than issuing an extra caution. This means accepting more false positives in exchange for a stronger safeguard against catastrophic harm. The ethical logic is simple but profound: inconvenience can be managed, but preventable exposure to contaminated water cannot be undone. In practice, this reflects a precautionary approach that places human life, public safety, and vulnerable populations ahead of efficiency or convenience.

Access Bias

There is a serious risk that only communities with reliable internet access, smartphones, or digital literacy can benefit from the system. If that happens, the very people most likely to face water contamination—often the poorest, most remote, and most marginalized communities—could be excluded from protection. That would turn a tool meant to reduce inequality into one that reinforces it. Ethical deployment therefore requires more than technical performance; it demands inclusive design, low-barrier access, multilingual support, and pathways for offline or community-based use so that the benefits of the system reach those who need them most.

Where Do We Go From Here?

IoT Sensor Integration

Connecting the model directly to smart water pipes for automated monitoring would move the system from periodic analysis to continuous surveillance. By pairing machine learning with distributed IoT sensors, the predictor could flag emerging contamination patterns in real time and notify water authorities immediately. That kind of rapid alerting creates an opportunity for intervention before unsafe water spreads through a network, protecting entire neighborhoods and reducing the scale of public health emergencies.

Mobile App Native Deployment

Creating an offline Android/iOS app for extremely remote areas without internet would make the tool far more accessible and resilient. Instead of depending on cloud connectivity, community members could run water safety checks locally on a phone, even in places with weak infrastructure or intermittent service. This would empower households, field workers, and local health teams in developing regions to make faster, independent decisions and extend the benefits of the model to populations that are often hardest to reach.

Adding Biological Data

Expanding the dataset to include proxy metrics for bacterial contamination would strengthen the model beyond its current chemical focus. At present, the system is well positioned to detect chemical hazards, but water safety is broader than that alone. Incorporating biological indicators would create a more complete assessment of risk, improving the model’s ability to identify multiple threat categories and supporting a more comprehensive, future-ready water quality monitoring framework.

Final Thoughts

Machine learning is a viable, powerful tool for environmental health monitoring. When designed responsibly and guided by ethical considerations, AI can address urgent real-world problems that affect millions of people globally, helping communities detect risks earlier and respond more effectively.

We successfully transitioned water testing from a delayed laboratory process to an instant predictive model. This shift demonstrates how technology can democratize access to critical safety information, reduce dependence on slow infrastructure, and empower communities with faster, more actionable insight.

Ethical AI design requires prioritizing human safety over pure statistical accuracy. The choices we make in model development have real consequences, and our responsibility is ultimately to the people whose lives depend on these systems, not just the numbers our models produce.

References & Resources

Data & Methodology

- Kaggle: Water Potability Dataset
- Scikit-Learn Documentation: Support Vector Machines and Classification Metrics
- Python Libraries: Pandas, NumPy, Scikit-Learn

Deployment & Technology

- Streamlit Documentation: Building Interactive Web Applications
- Joblib: Model Persistence and Serialization

Global Impact & Ethics

- United Nations Sustainable Development Goals (SDGs 3: Good Health and Well-being, Goal 6: Clean Water and Sanitation)
- WHO Guidelines on Drinking Water Quality
- IEEE Standards on Ethical AI in Critical Systems